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Sub ject : DSA

## Unit 2, S4, SLO1 - Array Implementation of Lists - Programming Practice 1

### 1. Find the max and min elements in an array.

**Answer:** #include <stdio.h>  
int main() {  
int a[] = {25, 11, 7, 75, 56}, n = 5;  
int max = a[0], min = a[0];  
for(int i = 1; i < n; i++) {  
if(a[i] > max) max = a[i];  
if(a[i] < min) min = a[i];  
}  
printf("Max: %d, Min: %d\n", max, min);  
return 0;  
}  
Output: Max: 75, Min: 7

### 2. Count frequency of each element in an array.

**Answer:** #include <stdio.h>  
int main() {  
int a[] = {1, 2, 8, 3, 2, 2, 8, 5}, n = 8, visited = -1;  
int fr[8];  
for(int i = 0; i < n; i++) {  
int count = 1;  
for(int j = i + 1; j < n; j++) {  
if(a[i] == a[j]) { count++; fr[j] = visited; }  
}  
if(fr[i] != visited) fr[i] = count;  
}  
for(int i = 0; i < n; i++) if(fr[i] != visited) printf("%d: %d times\n", a[i], fr[i]);  
return 0;  
}

### 3. Separate out the negatives, positives and zeroes.

**Answer:** #include <stdio.h>  
int main() {  
int a[] = {-4, 3, 0, -2, 5, 0, 7}, n = 7;  
printf("Positives: "); for(int i=0;i<n;i++) if(a[i]>0) printf("%d ", a[i]);  
printf("\nNegatives: "); for(int i=0;i<n;i++) if(a[i]<0) printf("%d ", a[i]);  
printf("\nZeroes: "); for(int i=0;i<n;i++) if(a[i]==0) printf("%d ", a[i]);  
return 0;  
}

### 4. Reverse the array elements.

**Answer:** #include <stdio.h>  
int main() {  
int a[] = {10, 20, 30, 40, 50}, n = 5;

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for(int i = 0; i < n / 2; i++) {  
    int temp = a[i]; a[i] = a[n - 1 - i]; a[n - 1 - i] = temp;  
}  
printf("Reversed: "); for(int i=0;i<n;i++) printf("%d ", a[i]);  
return 0;  
}
```

Output: Reversed: 50 40 30 20 10

#### 5. Print odd numbers in an array.

**Answer:** #include <stdio.h>  
int main() {  
 int a[] = {12, 17, 19, 24, 33, 40}, n = 6;  
 printf("Odd numbers: ");  
 for(int i = 0; i < n; i++) if(a[i] % 2 != 0) printf("%d ", a[i]);  
 return 0;  
}

Output: Odd numbers: 17 19 33